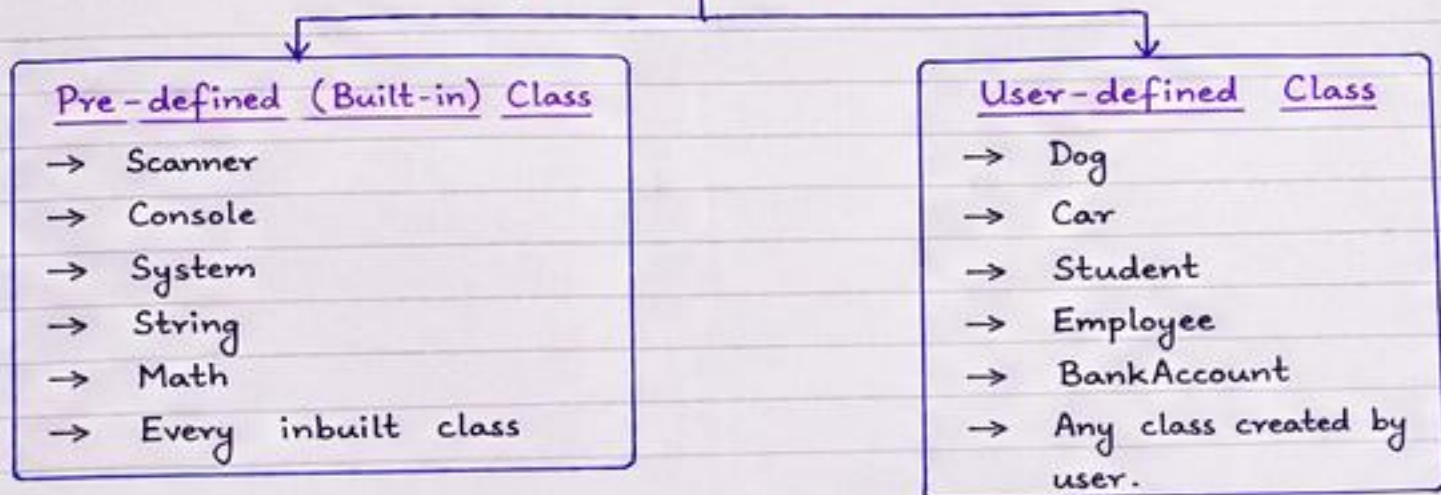


CLASS IN JAVA

Q. What is Class?

Ans → Class is a collection of objects and it doesn't take any space on memory. Class is also called as blueprint / logical entity.

TYPES OF CLASS



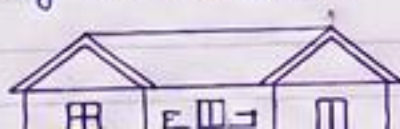
Key Idea :- Class is a template or blueprint for an object and object is the instance of the class.

Real Life Example :-

- Blueprint of a house (class) contains all the details like rooms, doors, windows.
- Actual house (object) is built using that blueprint.



→
Blueprint



USER-DEFINED CLASS

A class which is created by Java programmer is called user-defined class.

SYNTAX :-

```
class ClassName
{
    // data members (fields)
    dataType variable1;
    dataType variable2;
    .....

    // methods (functions)
    returnType methodName()
    {
        // code
    }
    .....
}
```

Example :-

```
Class Student
{
    int rollNo;
    String name;

    void show()
    {
        System.out.println("Hello");
    }
}
```

←→ Class Name

←→ Data Members (Variables)

→ Method

Points to Remember

1. Class is the collection of data and methods.
2. Methods defined inside the class are called members of the class.
3. In-built classes have Caps in first character but in user-defined classes it is not mandatory.
4. Class is a logical entity and Object is the real world entity.
5. In user-defined class, functions are always user-defined.

CLASS vs OBJECT

Class (Logical Entity)

- It is a blueprint or template.
- It does not occupy memory.
- It is declared using 'class' keyword.
- It defines properties and behaviors.

Object (Real World Entity)

- It is an instance of class.
- It occupies memory.
- It is created using 'new' keyword.
- It represents real world things.

Example :-

Class

```
Class Car
{
    String color;
    int speed;
    void display()
    {
        System.out.println("Car Details");
    }
}
```

Creates →

Objects

```
Car c1 = new Car();
c1.color = "Red";
c1.speed = 60;
```



Object 1

```
Car c1 = new Car();
c1.color = "Blue";
c1.speed = 80;
```



Object 2

Real Life Example :-

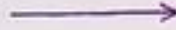
- Class is like a general recipe of a dish.
- Objects are actual dishes prepared using that recipe.



Recipe
(Class)



Dish 1
(Object)



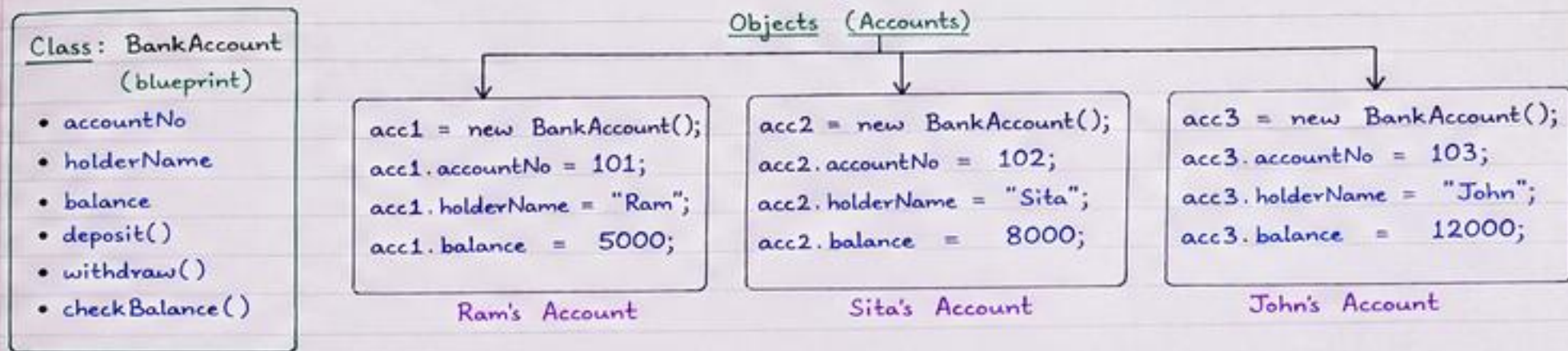
Dish 2
(Object)

IMPORTANCE OF CLASS

1. Helps in Code Reusability — code can be reused by creating objects.
2. Improves Modularity — programs are divided into different classes.
3. Easy Maintenance — changes in one class do not affect other classes.
4. Security — data hiding can be achieved using classes.
5. Real world modeling — classes represent real world entities.

MORE REAL LIFE EXAMPLE :-

Consider a 'Bank Account' system.



Summary :- Class is the blueprint (logical) and Object is the instance (real).
Class defines what data and behavior an object will have.